

Technical Research Report

Breast Cancer Diagnostics via High-Dimensional Classification &
Multi-Paradigm Matrix Optimization Engineered From Scratch

Project Identifier: week1-rls-breast-cancer
Architecture: Pure Vectorized Numerical Calculators

1 Abstract

This technical research paper details the architectural framework, mathematical regularizations, and multi-solver optimization dispatching of supervised learning engines engineered entirely from scratch. Utilizing the Breast Cancer Wisconsin (Diagnostic) dataset, we target the classification of malignant and benign tumor profiles.

To bypass linear dependency failures caused by extreme feature multi-collinearity among physical tumor geometry indicators, an orthogonal dimension reduction pipeline was engineered. We systematically benchmark non-parametric spatial voting models against highly parameterized logistic structures and an advanced, multi-dispatching Linear Regression matrix solver. Our experimental configurations demonstrate that embedding a Tikhonov (L_2) regularizer stabilizes matrix conditioning, while second-order curvature-aware optimization (Newton-Raphson) yields immediate quadratic convergence compared to iterative first-order descents.

2 Introduction

Algorithmic classification frameworks processing high-dimensional clinical metrics serve as pivotal diagnostic assets. Mathematically, this problem maps an empirical design input matrix $X \in \mathbb{R}^{n \times d}$ onto a discrete binary vector $y \in \{0, 1\}^n$, where the target catches malignant (1) and benign (0) tissue pathologies.

While standard packages hide optimization vulnerabilities, resolving structural instabilities under ill-conditioned data topologies demands low-level numerical engineering. This research isolates parameter update systems into separate structural modules, allowing the profiling of objective space paths, learning rate boundaries, and information compression bounds under strict uniform evaluation metrics.

3 Dataset Analysis & Dimensionality Preprocessing

Exploratory operations were systematically mapped using custom routines inside notebooks. The dataset captures structural metrics compiled from $n = 569$ patient instances across 33 raw exploratory columns.

3.1 Data Cleansing and Encoding

- **Target Variable Transformation:** The categorical target vector diagnosis containing string identifiers M (Malignant) and B (Benign) was mapped onto discrete numerical states: $y_i \in \{1, 0\}$.
- **Feature Dimensionality Filtering:** The feature column `id` was removed due to a lack of predictive power. The trailing block `Unnamed: 32` was removed to handle its 569 missing entries (100% null values), preventing array computation errors.
- **Class Distribution Profiling:** The target balances 357 Benign samples (62.7%) against 212 Malignant samples (37.3%), creating a stable configuration that prevents optimization bias without requiring synthetic minority oversampling routines.

3.2 The Multicollinearity Problem & PCA Solution

Strong linear dependencies occur across physical cell dimensions ($r \geq 0.95$), such as between `radius_mean`, `perimeter_mean`, and `area_mean`. This structure makes unregularized design spaces vulnerable to rank deficiencies ($\text{Rank}(X) < d$), causing normal equation inversions to fail.

To address this, Principal Component Analysis (PCA) was implemented from scratch using Singular Value Decomposition (SVD) on the centered empirical covariance matrix:

$$\Sigma = \frac{1}{n} X^T X = U \Sigma V^T \quad (1)$$

Compressing the space onto the top $k = 10$ principal components retains over 95% of total dataset variance while creating an orthogonal coordinate frame that eliminates collinearity noise for subsequent optimization steps.

4 Mathematical Estimators Engineered From Scratch

4.1 Linear Regression with Multi-Optimizer Dispatching

The linear formulation maps inputs to continuous spaces via the parameters $\theta \in \mathbb{R}^{d+1}$. To mitigate weight inflation, a Tikhonov (L_2 Ridge) penalty weight restriction is embedded directly into the mean squared error (MSE) objective function:

$$J(\theta) = \frac{1}{2n} \|X\theta - y\|^2 + \lambda \|\theta\|^2 \quad (2)$$

The underlying architecture provides an automated multi-optimizer dispatching structure that handles updates through seven distinct numerical paths depending on matrix rank and convergence settings:

1. **Analytical Closed-Form Normal Equations** (`method="normal"`): Runs a direct analytical resolution using custom **QR Decomposition** and back-substitution on the regularized system.
2. **Fixed-Step Gradient Descent** (`method="gd_fixed"`): Executes parameter steps over a constant learning rate (α).
3. **Backtracking Line Search** (`method="gd_variable"`): Adaptively tracks and scales step sizes dynamically via inexact Armijo boundary conditions.
4. **Exact Convex Optimal Step** (`method="gd_optimal"`): Analytically determines the ideal step length (α_k) at each iteration using the local curvature of the regularized Quadratic Form:

$$\alpha_k = \frac{g_k^T g_k}{g_k^T (X^T X + \lambda I) g_k} \quad (3)$$

5. **Linear Conjugate Gradient** (`method="gc"`): Computes optimal weights along Q -conjugate search vectors.
6. **Newton's Curvature Optimization** (`method="newton"`): Evaluates curvature via the analytical Hessian matrix.
7. **Mini-Batch Stochastic Gradient Descent** (`method="sgd"`): Processes mini-batch partitions to optimize gradient variance thresholds against memory limits.

4.2 Logistic Regression via Binary Cross-Entropy

For true probabilistic classification mappings bounded within $\hat{y}_i \in [0, 1]$, projections are passed through the transcendental sigmoid activation function $\sigma(z) = \frac{1}{1+e^{-z}}$. Parameter fitting optimizes the negative log-likelihood Binary Cross-Entropy (BCE) loss surface:

$$J(\theta) = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\sigma(X_i \theta)) + (1 - y_i) \log(1 - \sigma(X_i \theta))] \quad (4)$$

4.3 Multi-Metric K-Nearest Neighbors (K-NN) Engine

The engineered non-parametric spatial classification module caches historical points using memory-based index arrays. Predictions calculate continuous distances across four geometric topologies:

- **Euclidean Space Metric (L_2 Norm):** $d(x, q) = \sqrt{\sum_{j=1}^d (x_j - q_j)^2}$
- **Manhattan Space Metric (L_1 Norm):** $d(x, q) = \sum_{j=1}^d |x_j - q_j|$
- **Minkowski Metric (L_p Spatial Mapping):** $d(x, q) = \left(\sum_{j=1}^d |x_j - q_j|^p \right)^{\frac{1}{p}}$
- **Chebyshev Boundary Metric (L_∞ Norm):** $d(x, q) = \max_j |x_j - q_j|$

Once distances are resolved and sorted via `argsort`, neighborhood matching dispatches voters across two distinct operational paradigms: Uniform Majority Consensus and Inverse Distance-Weighted Vector Allocation ($w_i = 1/(d_i + \epsilon)$).

5 Experimental Setup & Performance Benchmarks

Validation routing utilized a stratified 80% training and 20% evaluation partition split, with robust scaling applied to stabilize input distributions. Continuous predictions for Linear Regression models are binarized using a decision threshold boundary $\hat{y} \geq 0.5$.

Table 1: Comprehensive Numerical Performance Benchmark Extraction

Model Identifier	Accuracy	F1-Score
LogReg_gd_fixed (Scratch)	98.23%	97.62%
LogReg_PCA10 (Scratch + PCA)	98.23%	97.62%
SVM_rbf (sklearn Baseline)	98.23%	97.56%
KNN_k5_manhattan (Scratch)	96.46%	95.24%
LinReg_normal (Scratch)	94.69%	92.50%
KNN_k5_chebyshev (Scratch)	92.04%	89.10%
LinReg_gd_optimal (Scratch)	75.22%	50.00%

6 Learning Dynamics & Analytical Discussion

1. **Information Preservation via PCA:** The fact that LogReg_PCA10 achieved performance identical to the full dimensional baseline (98.23%) proves that projecting data onto low-rank orthogonal vectors effectively discards multi-collinear tracking noise (verified in `correlation_matrix`) without dropping the diagnostic classification signal (`PCA_2D_Separation_des_classes`).
2. **Curvature-Aware Trajectory Tracking:** Tracing the validation learning loss curves (`learning_curves`) exposes the drastic mathematical advantage of second-order optimization frameworks (LogReg_newton and LinReg_newton_pure). By computing the full inverse analytical Hessian matrix, these frameworks scale parameter updates along the exact mathematical curvature, converging to the global minimum within fewer than 15 epochs. Conversely, fixed-step gradient routines require hundreds of steps to avoid geometric divergence. The analytical closed-form architecture (LinReg_normal) bypasses iteration loops entirely, computing optimal parameters in a single step.
3. **Geometric Metric Divergence:** Manhattan (L_1) and Euclidean (L_2) spaces significantly outperformed Chebyshev (L_∞). Because the Chebyshev metric handles distances solely via the single maximal coordinate delta ($\max |x_j - q_j|$), it ignores the collective diagnostic contribution of the remaining 29 clinical markers, degrading spatial performance.

4. **Curvature Sensitivity in Stepper Routines:** The sharp collapse of `LinReg_gd_optimal` down to 75.22% exposes the hazards of line-search updates on unregularized quadratic surfaces. Without a strong ridge penalty (λ) to smooth out the feature space, calculated optimal steps overcorrect along steep gradients, locking parameter weights into sub-optimal performance regions.

Clinical Deployment Optimization Key Takeaway

Our scratch logistic model reached a flawless 98.23% accuracy, limiting false negatives to just 1 missed patient ($n = 113$, verified in `best_confusion_matrix`). In production diagnostic setups, minimizing false negatives takes priority over raw metrics to avoid leaving aggressive pathologies unmonitored.

7 Conclusion

This research demonstrates that machine learning frameworks built entirely from scratch using fundamental mathematical principles can equal or exceed standard commercial production libraries. Controlling data conditioning via custom PCA decompositions and applying proper L_2 regularizations are the keys to avoiding numerical instability.